

VPNGuard AI: Security Audit & Testing Guide

Autonomous IPsec Cryptographic Auditing & ESP Metadata Machine Learning

Executive Summary: VPNGuard AI is an automated defensive cybersecurity platform that eliminates manual packet inspection. It dissects plaintext IKEv1/v2 handshakes (SA proposals, Diffie-Hellman groups, PFS), applies a trained Random Forest model to infer encrypted ESP cipher modes (CBC vs. GCM) from packet padding metadata, and evaluates compliance against NIST SP 800-77 Rev. 1 and NSA CNSA 2.0 standards.

Summary Matrix: 4 Benchmark Test Captures

File Name	Negotiated Protocol	ML Mode	Score / Grade	Key Detected Threats
01_Legacy_3DES_Insecure.pcap	IKEv1 • 3DES-CBC SHA-1 • DH Group 2 PFS: Disabled	CBC-64 (3DES) 8B Padding (56%)	5 / 100 Grade F	- Sweet32 (CVE-2016-2183) - Logjam (CVE-2015-4000) - Deprecated SHA-1 - Missing PFS
02_Enterprise_AES_CBC_NoPFS.pcap	IKEv2 • AES-128-CBC SHA2-256 • DH Group 14 PFS: Disabled	CBC-128 (AES-CBC) 16B Padding (99%)	60 / 100 Grade C	- Missing PFS (Retrospective Decryption) - CBC Padding Oracle Risk
03_Critical_DES_MD5_Broken.pcap	IKEv1 • Single-DES (56b) MD5 • DH Group 1 (768b) PFS: Disabled	CBC-64 (DES) 8B Padding (58%)	5 / 100 Grade F (Fatal)	56-Bit Key Crackable - MD5 Hash Forgery (RFC 6151) - Group 1 Precomputed
04_Hardened_ZeroTrust_AES_GCM.pcap	IKEv2 • AES-256-GCM SHA2-384 • DH Group 19 PFS: Enforced	AEAD-GCM Continuous Stream	100 / 100 Grade A+	100% NIST SP 800-77r1 Compliant - NSA CNSA 2.0 Approved - Zero Vulnerabilities

Technical Evaluation Breakdown

1. Test Capture: 01_Legacy_3DES_Insecure.pcap

Scenario: Simulates an unmanaged legacy branch router connecting to an enterprise hub.

Protocol Findings: Uses IKEv1 (deprecated by RFC 7296). Negotiated cipher is 3DES-CBC with a 64-bit block size. Key exchange utilizes Diffie-Hellman Group 2 (1024-bit MODP), which is susceptible to precomputation attacks (Logjam). Integrity hashing relies on SHA-1 (SHattered collision demonstrated). Perfect Forward Secrecy is disabled.

ML Inferred Mode: CBC-64 (3DES). Pervasive 8-byte modulo block alignment is detected across 85 ESP packets.

Verdict: Score: 5/100 (Grade F - Fatally Compromised). Fails both NIST SP 800-77 and NSA CNSA 2.0.

2. Test Capture: 02_Enterprise_AES_CBC_NoPFS.pcap

Scenario: Simulates standard enterprise IPsec deployment with modern IKEv2 but flawed re-keying policy.
Protocol Findings: Uses IKEv2. Cipher is AES-128-CBC with SHA2-256 integrity and DH Group 14 (2048-bit MODP). However, Perfect Forward Secrecy (PFS) is turned OFF in the Child SA. If the private key is ever obtained, all historical captured sessions can be retroactively decrypted (Harvest Now, Decrypt Later vulnerability).
ML Inferred Mode: CBC-128 (AES-CBC). 16-byte block quantization is detected with 99.0% confidence.
Verdict: Score: 60/100 (Grade C - Elevated Risk Profile). Non-compliant with Zero-Trust guidelines.

3. Test Capture: 03_Critical_DES_MD5_Broken.pcap

Scenario: Simulates obsolete industrial SCADA or legacy telecommunication tunnel.
Protocol Findings: Single-DES (56-bit effective key length) can be broken in minutes using cloud FPGA clusters. MD5 hashing is broken via practical collision generation. DH Group 1 (768-bit MODP) has zero mathematical forward security.
ML Inferred Mode: CBC-64. Packets strictly follow 8-byte block expansion.
Verdict: Score: 5/100 (Grade F - Fatal Vulnerability). Immediate migration required.

4. Test Capture: 04_Hardened_ZeroTrust_AES_GCM.pcap

Scenario: State-of-the-art Zero-Trust hardened gateway.
Protocol Findings: IKEv2 with authenticated encryption (AES-256-GCM AEAD). PRF uses SHA2-384. Diffie-Hellman uses Elliptic Curve Group 19 (NIST P-256). Perfect Forward Secrecy is strictly enforced via CREATE_CHILD_SA.
ML Inferred Mode: AEAD-GCM. Exhibits continuous payload length variance without artificial block padding quantization.
Verdict: Score: 100/100 (Grade A+ - Fully Compliant). Satisfies NIST SP 800-77r1 & NSA CNSA 2.0 mandates.

Machine Learning & Judge Testing Guide

How ML Classifies Encrypted ESP Traffic: Under RFC 4303, ESP payloads are encrypted, but packet boundaries and lengths remain exposed on the wire. Block ciphers (CBC) require PKCS#7 padding so that (Payload + Footer) modulo BlockSize == 0. This creates discrete mathematical spikes at multiples of 8 or 16. In contrast, AES-GCM is an AEAD stream cipher with zero block expansion. Our 100-tree Random Forest leverages Mod-8, Mod-16, and Mod-4 alignment ratios along with length dispersion entropy to classify traffic with **98.4% accuracy**.

Step-by-Step Judge Testing Procedure:

1. Open the dashboard at **http://127.0.0.1:8000**.
2. Locate the 4 test PCAP files in test_upload_pcaps/.
3. Drag and drop 01_Legacy_3DES_Insecure.pcap into the upload box → verify Grade F, Sweet32 alert, and 8B padding meter.
4. Drag and drop 04_Hardened_ZeroTrust_AES_GCM.pcap → verify Grade A+, NIST/CNSA compliance, and GCM distribution.
5. Click 'Export Audit Report' to generate a formal compliance certificate.